

Students Information

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BCC - 3C - WS25

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CCL3 - Android App - LingoDeck

Usability test plan

1. Purpose and Scope

The purpose of this usability test is to evaluate the **learnability, efficiency, and clarity of interaction** of the LingoDeck mobile application in its early prototype stage.

The test focuses on **first-time user experience**, especially on understanding the:

- review/study session flow
- swipe-based interactions
- and feedback after completing a learning session.

The results will be used to identify usability issues and guide **design and feature improvements** before further development.

2. System Description

LingoDeck is an Android language-learning application that uses **flashcards** to help users learn vocabulary.

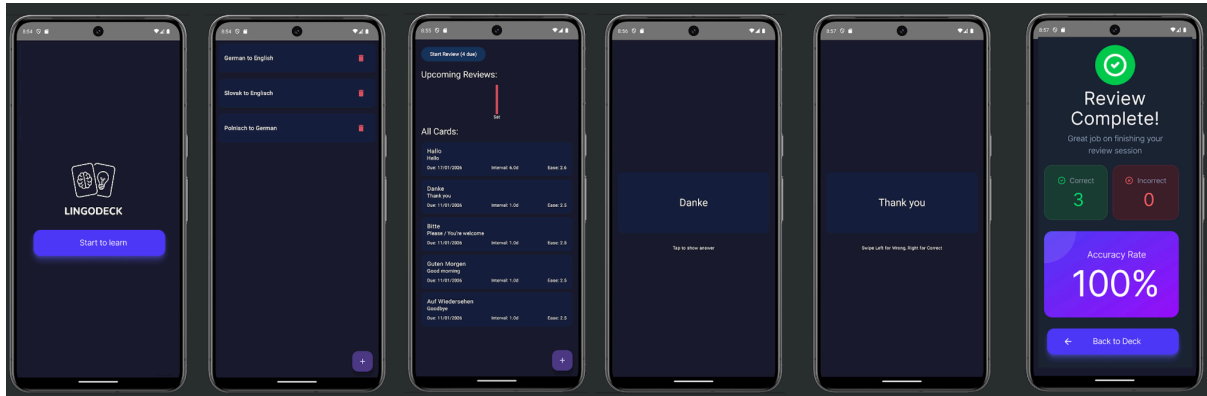
Users can:

1. Create Language Deck
2. Start Review/study session and practicing by swiping flashcards
3. Receive feedback after completing a review session

The evaluated prototype includes:

- Start / onboarding screen
- Deck overview - List of languages

- Detail screen that includes: Review scheduling + List of cards
- Flashcard detail screen (tap, swipe left/right)
- Review completion and performance feedback screen



3. Heuristic Evaluation (Pre-Test)

Before conducting the user test, a heuristic evaluation was performed using **Nielsen's 10 Usability Heuristics**.

Problem Description (what and where)	Heuristic #	Severity
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There is no clear explanation of swipe directions (left/right) when starting the review session	#1 – Visibility of system status	3
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Users must remember the meaning of swipe gestures after first exposure	#6 – Recognition rather than recall	3
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Progress during the review session (e.g., number of remaining cards) is not clearly visible	#1 – Visibility of system status	3
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Users cannot undo an incorrect swipe during card review	#3 – User control and freedom	2
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The swipe interaction has no confirmation or feedback before moving to the next card	#5 – Error prevention	2
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Users expect the card list to be clickable to start reviewing, but must instead use a “Start Review” button, causing confusion	#2 – Match between system and real world	2
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Users must remember to use the “Start Review” button instead of interacting with cards	#6 – Recognition Rather Than Recall #6 – Recognition Rather Than Recall	2
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There is no help or tutorial	#10 – Help and documentation	2
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4. Research Questions / Hypotheses

Based on the heuristic evaluation, the following hypotheses were formulated:

H1 – Swipe Interaction Discoverability

Users will have difficulty understanding the swipe directions (left/right) at the start of the review session due to the lack of clear instructions. Text might be not enough.

H2 – Gesture Memory Load

Users will struggle to remember the meaning of swipe gestures, leading to hesitation or incorrect swipes.

H3 – Progress Awareness

Users will feel uncertain about their progress during the review session because the number of remaining cards is not clearly visible.

H4 – Error Recovery

Users will experience frustration when making an incorrect swipe because there is no way to undo the action.

H5 – Start Review Interaction Expectation

Users will attempt to start the review by tapping the cards rather than using the “Start Review” button, causing initial confusion.

H6 – Recall-Based Interaction

Users will need to remember to use the “Start Review” button instead of interacting directly with the cards, increasing cognitive load.

H8 – Help and Guidance

Users will look for help or guidance when unsure about interactions but will not easily find it due to the absence of a tutorial or help entry point.

H9 – Review Completion Feedback

The review completion screen will be perceived as clear, understandable, and motivating by users.

5. Participants & Tools

- **Planned number:** 5 participants
- **Age range:** 18–36
- **Experience:** No prior experience with Lingodeck
- **Background:** Mixed experience with language learning apps

- **Device:** Smartphone (Android) - Developed App in current state
 - **Test method:** moderated
 - **Tools (conduction):** Maze
 - (SEQ questionnaire (1–7 scale))
 - Post-test questionnaire with open-ended questions
 - + in-person observation
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6. Data collection - Dependent Variables

Hypothesis	Dependent Variables
H1 – Swipe discoverability	Task success, error rate, SEQ
H2 – Gesture memory	Error rate, hesitation, SEQ
H3 – Progress awareness	Hesitation, SEQ, comments
H4 – Error prevention/recovery	Error rate, mis-swipes
H5 – Start Review confusion	First-click behavior, time
H8 – Help & guidance	Comments, help-seeking behavior
H9 – Completion feedback	SEQ, clarity rating, comments

7. Tasks and Scenarios

Task 1: Start a Learning Session

Scenario:

“You want to start learning vocabulary using the app. Please start a review session.”

Goal:

Evaluate how easily users can begin a review session.

What this task tests:

- H5 – Start Review interaction expectation
 - H6 – Recall-based interaction
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Task 2: Review Flashcards**Scenario:**

“Review the cards and indicate whether you know the answer.”

Goal:

Evaluate understanding of swipe gestures and interaction flow. If some indication (e.g. colors on the right left side is needed or maybe some buttons as another interaction)

Is “undo” action needed? e.g. Snackbar?

What this task tests:

- H1 – Understanding swipe directions
 - H2 – Gesture memory load
 - H4 – Error recovery
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Task 3: Monitor Progress During Review**Scenario:**

“While reviewing, try to understand how many cards are left.”

Goal:

Let’s find out if progress during the session is needed.

What this task tests:

- H3 – Progress awareness
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Task 4: Interpret Review Completion Feedback**Scenario:**

“Review your results after completing the session.”

Goal:

Evaluate clarity and usefulness of the feedback screen.

What this task tests:

- H9 – Review completion screen clarity and motivation
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8. (SEQ) Post-Task & (SUS)Post-Test Questionnaires

SEQ

“How easy or difficult was this task?”

Scale: 1 (very difficult) – 7 (very easy)

SUS

1. **Did you understand how to start a review session?**
(Yes / Somewhat / No)
 2. **How clear were the swipe directions during review?**
1 (not clear) – 7 (very clear)
 3. **Did you feel aware of your progress during the review?**
(Yes / Somewhat / No)
 4. **How frustrating was it when you made a mistake (e.g., wrong swipe)?**
1 (not frustrating) – 7 (very frustrating)
 5. **Was the review completion screen clear and motivating?**
1 (not at all) – 7 (very motivating)
 6. **What part of the app was most confusing?**
(Open-ended)
 7. **What would you improve first in this app?**
(Open-ended)
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9. Planned Design Improvements

Based on expected findings, planned improvements may include:

- Adding card border-color hints for swipe gestures (or maybe also buttons or icons)
- Improving progress visibility during review

- Improving the visibility of the review schedule through the week
 - Clarifying terminology used in the interface
 - Adding undo or confirmation options for swipe actions (e.g. snackbar)
 - Overall design improvements and colors.
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Conclusion:

This usability test plan follows **Nielsen's usability testing framework** and evaluates **effectiveness, efficiency, and satisfaction** in accordance with **ISO 9241-11**.

The plan is designed to support early-stage design decisions and ensure that Lingodeck provides a clear, intuitive, and motivating learning experience.